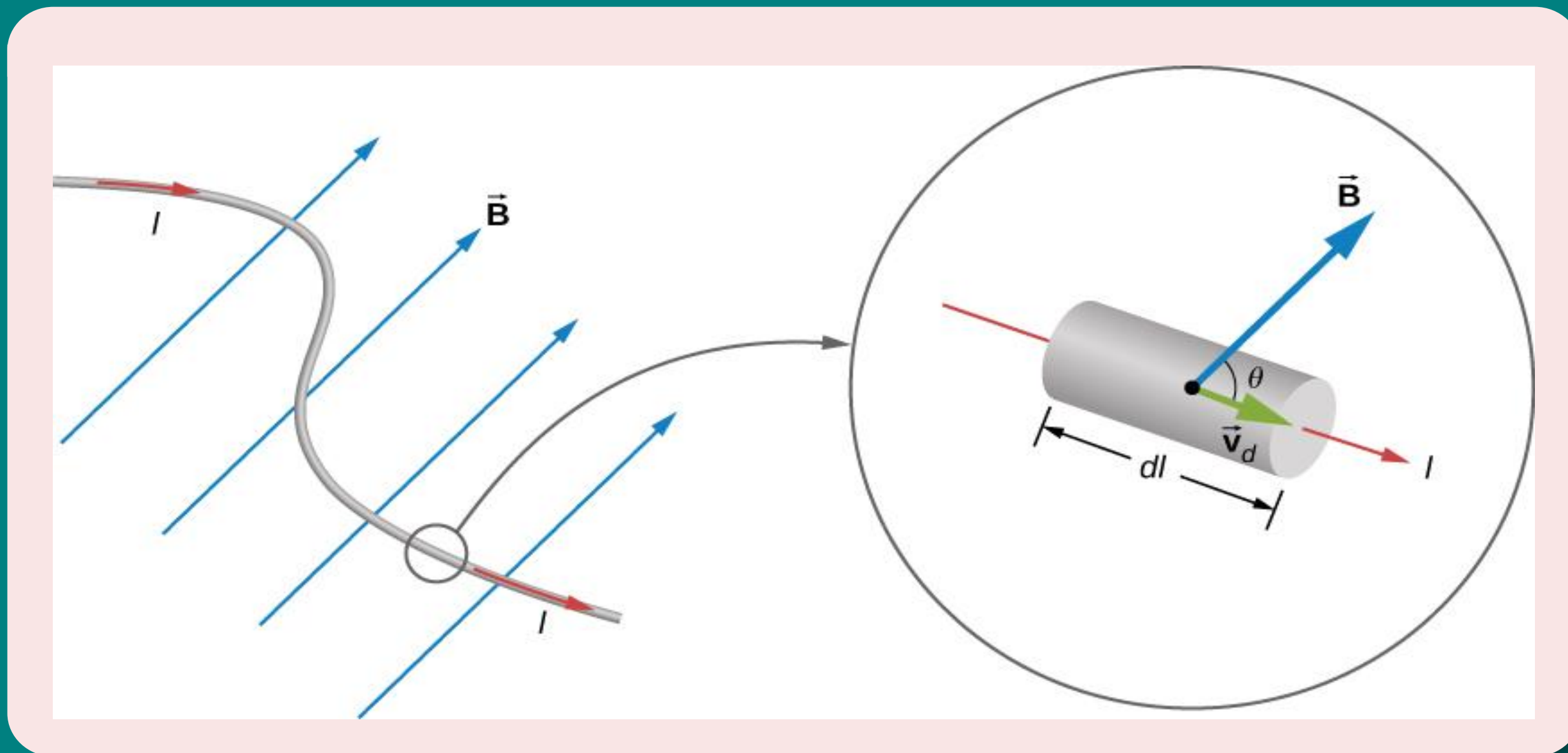
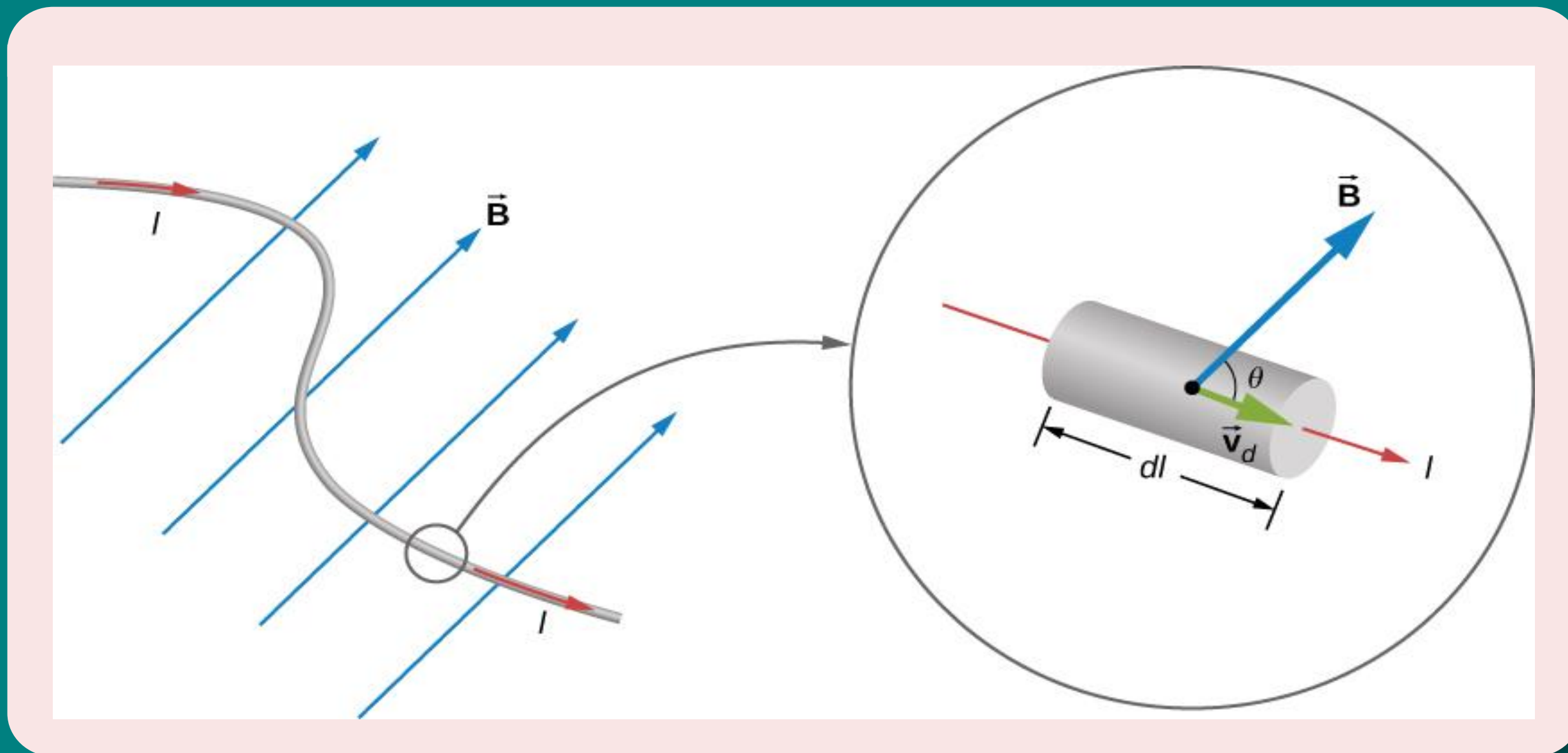


Magnetic force on a current-carrying conductor

- Electric current is **flow of charge**. A **current - carrying wire** in a **magnetic field** ($\vec{B}$) must therefore **experience a force** due to the field.

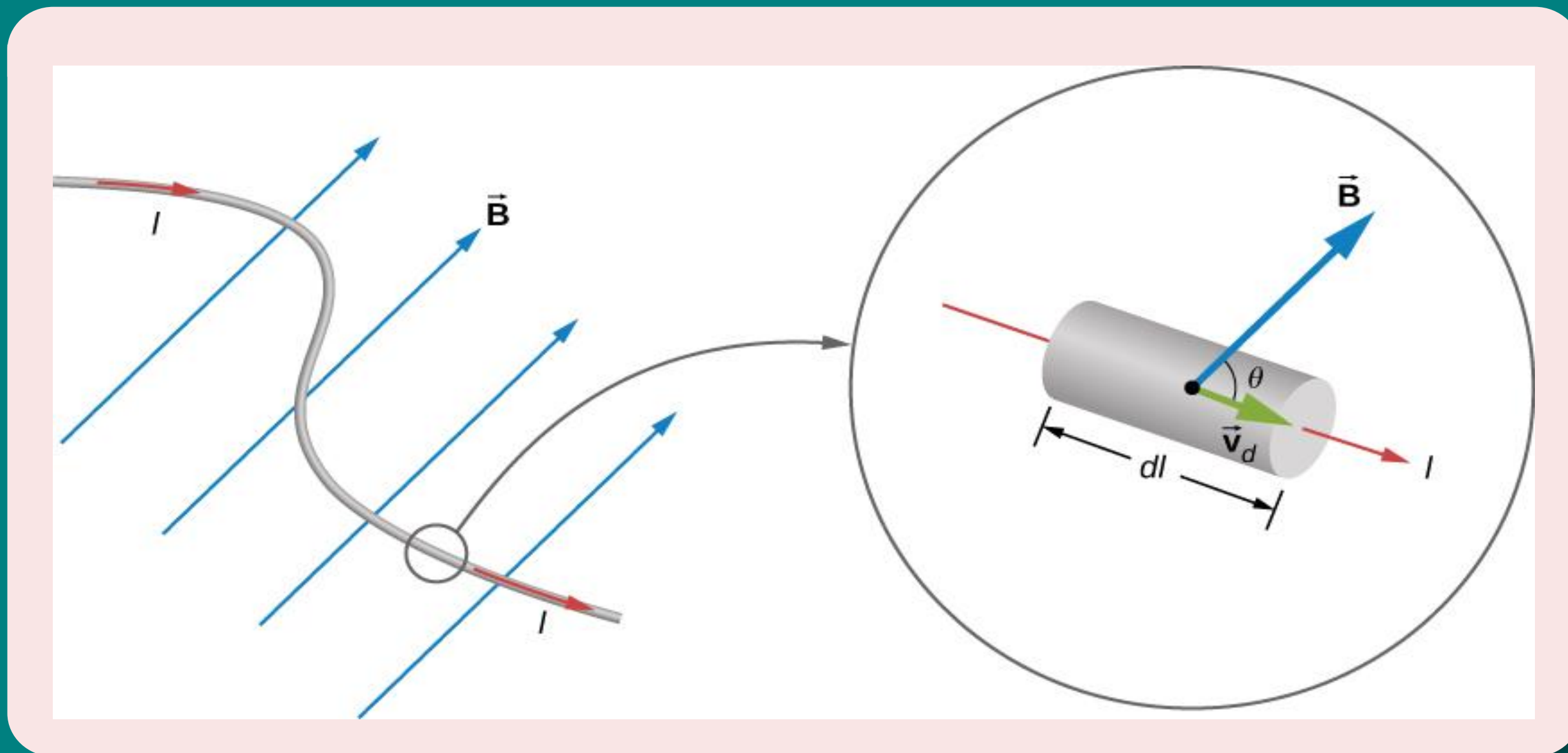


Magnetic force on a current-carrying conductor



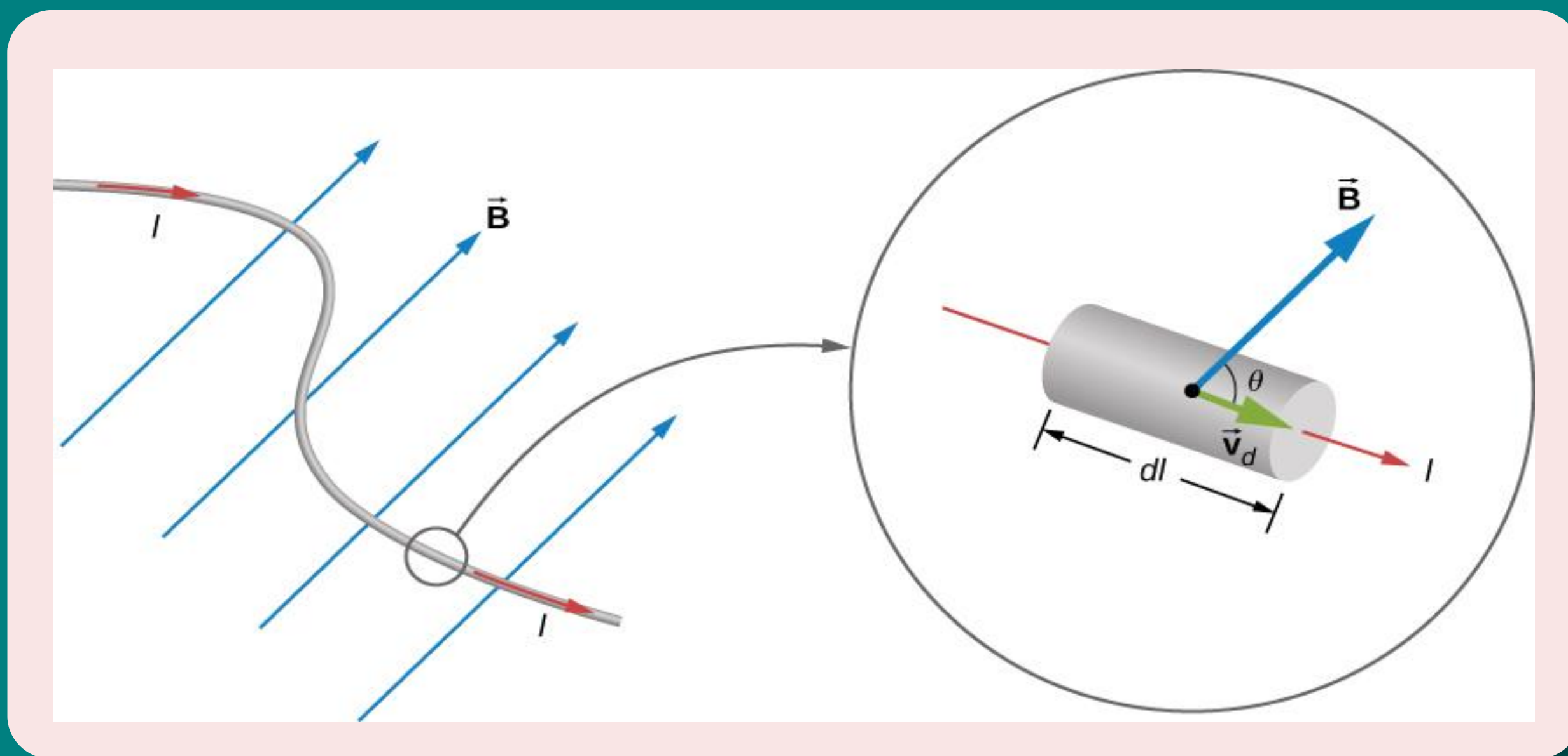
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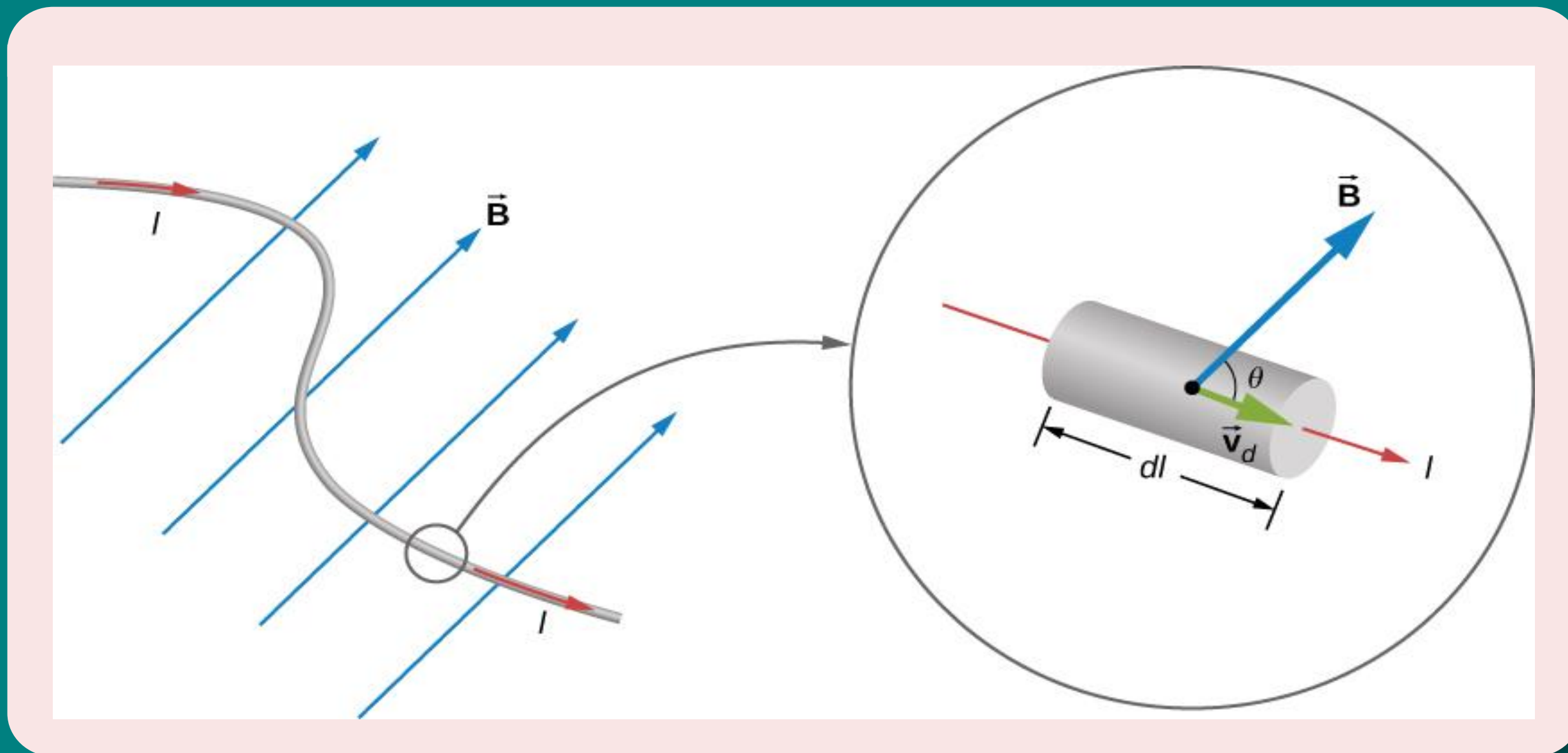
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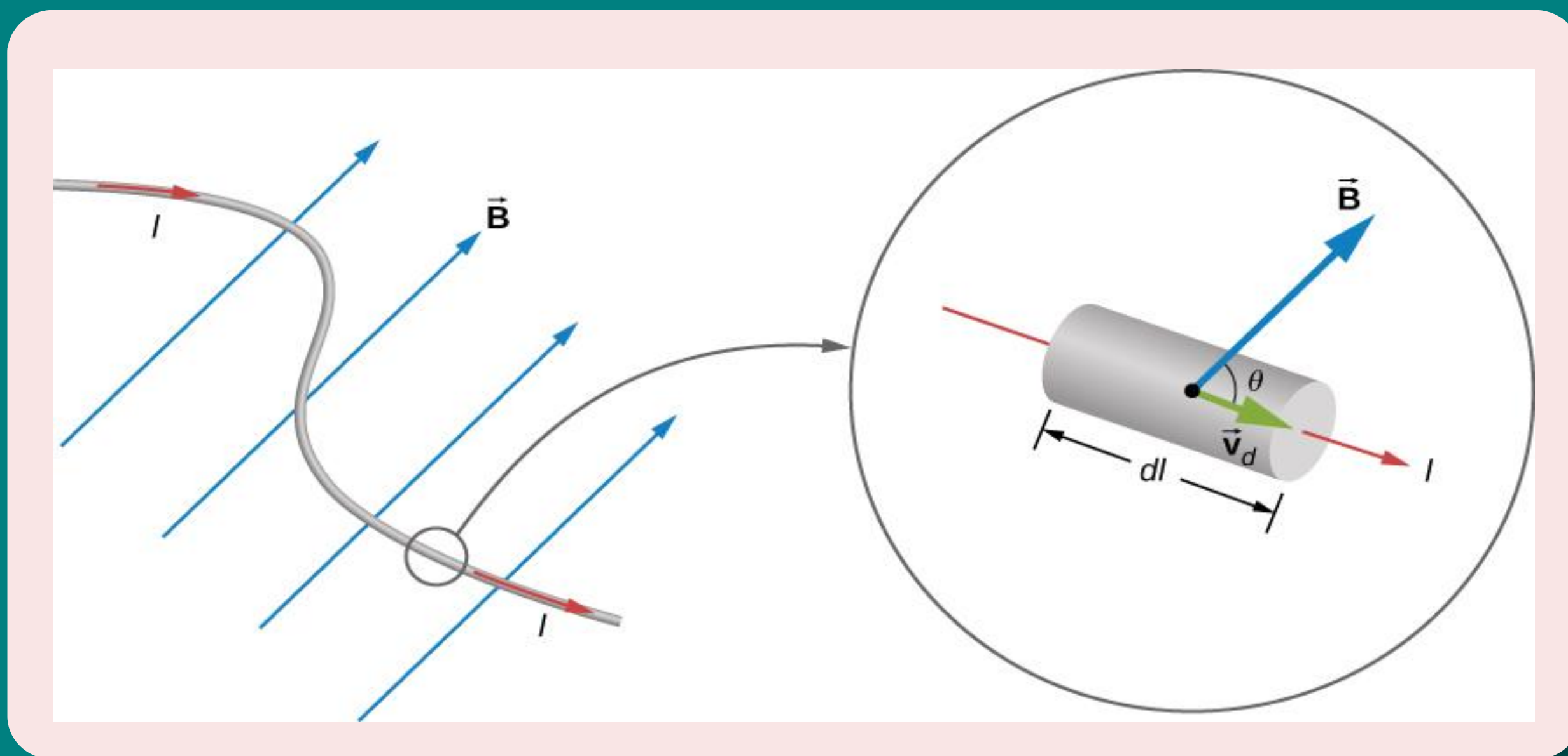
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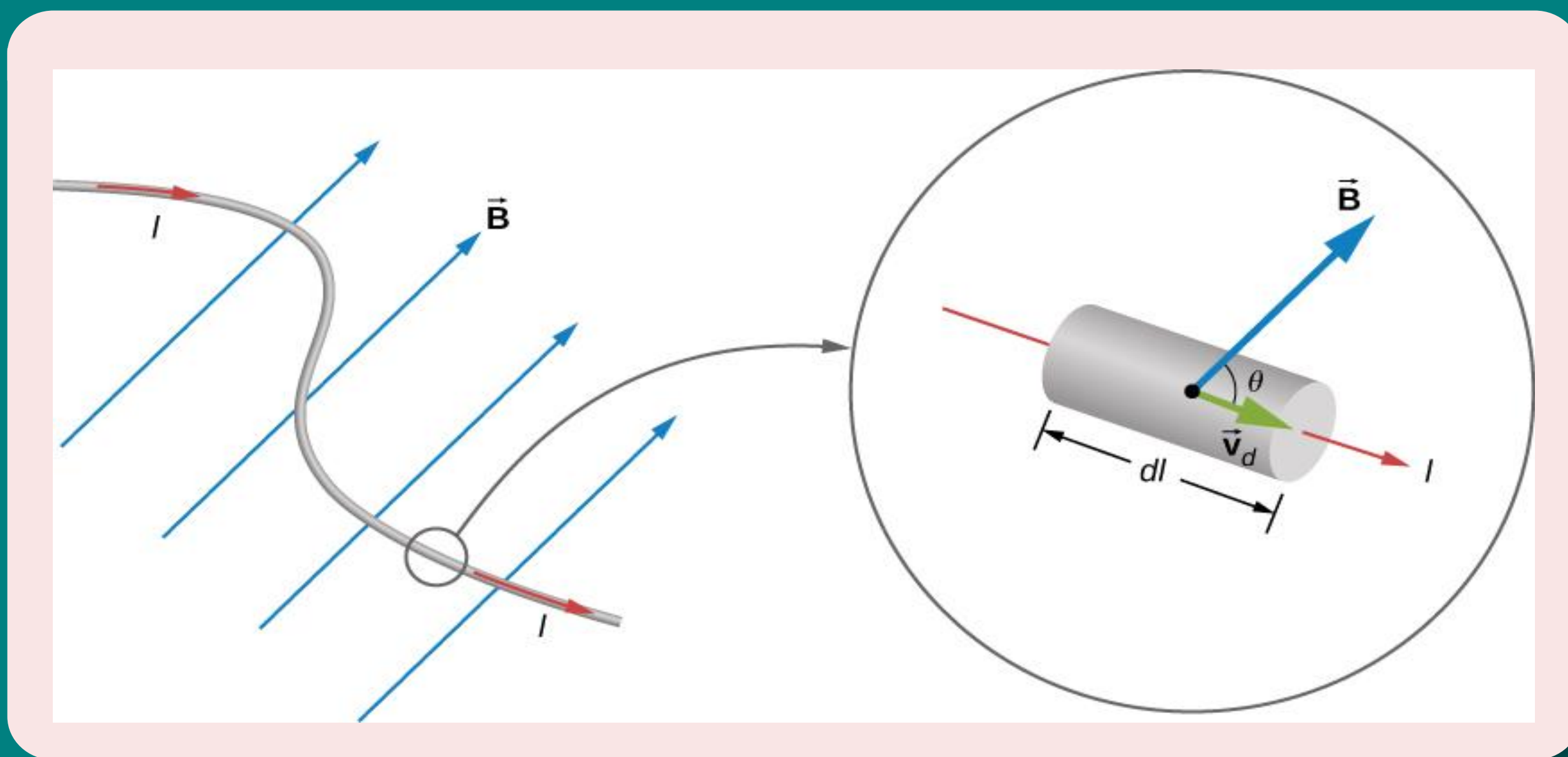
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Magnetic force on a current-carrying conductor



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- Let, n be the **charge carriers per unit volume**.
- So, the **number of charge carriers in the section** is $nV = nAdl$
- If the **charge carriers** move with **drift velocity** $\vec{v}_d$

Magnetic force on a current-carrying conductor

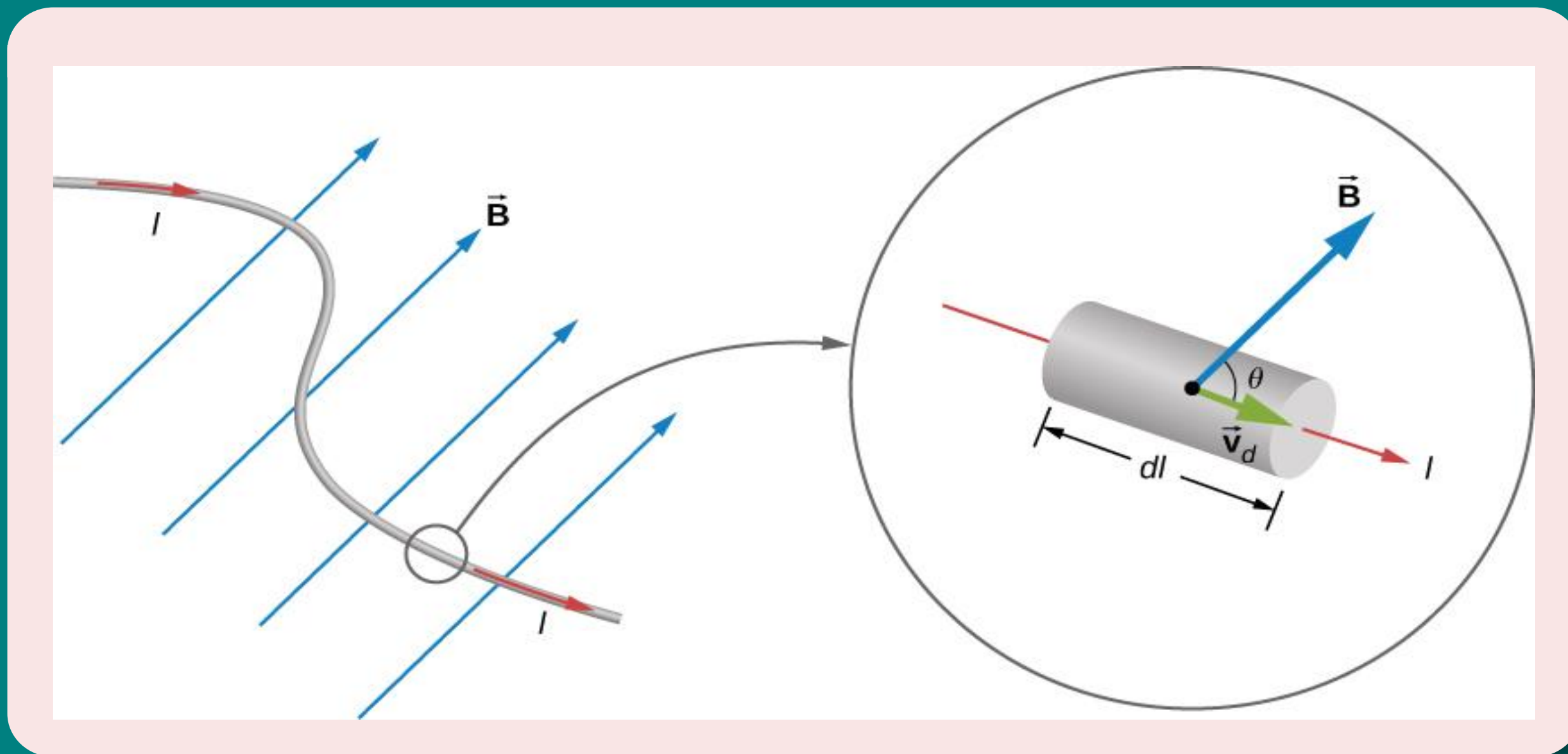


- **Force on the charge carriers** moving with velocity $\vec{v}_d$ in the section of wire is

$$d\vec{F} = Q\vec{v}_d \times \vec{B}$$
$$d\vec{F} = neAdl\vec{v}_d \times \vec{B}$$

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Magnetic force on a current-carrying conductor



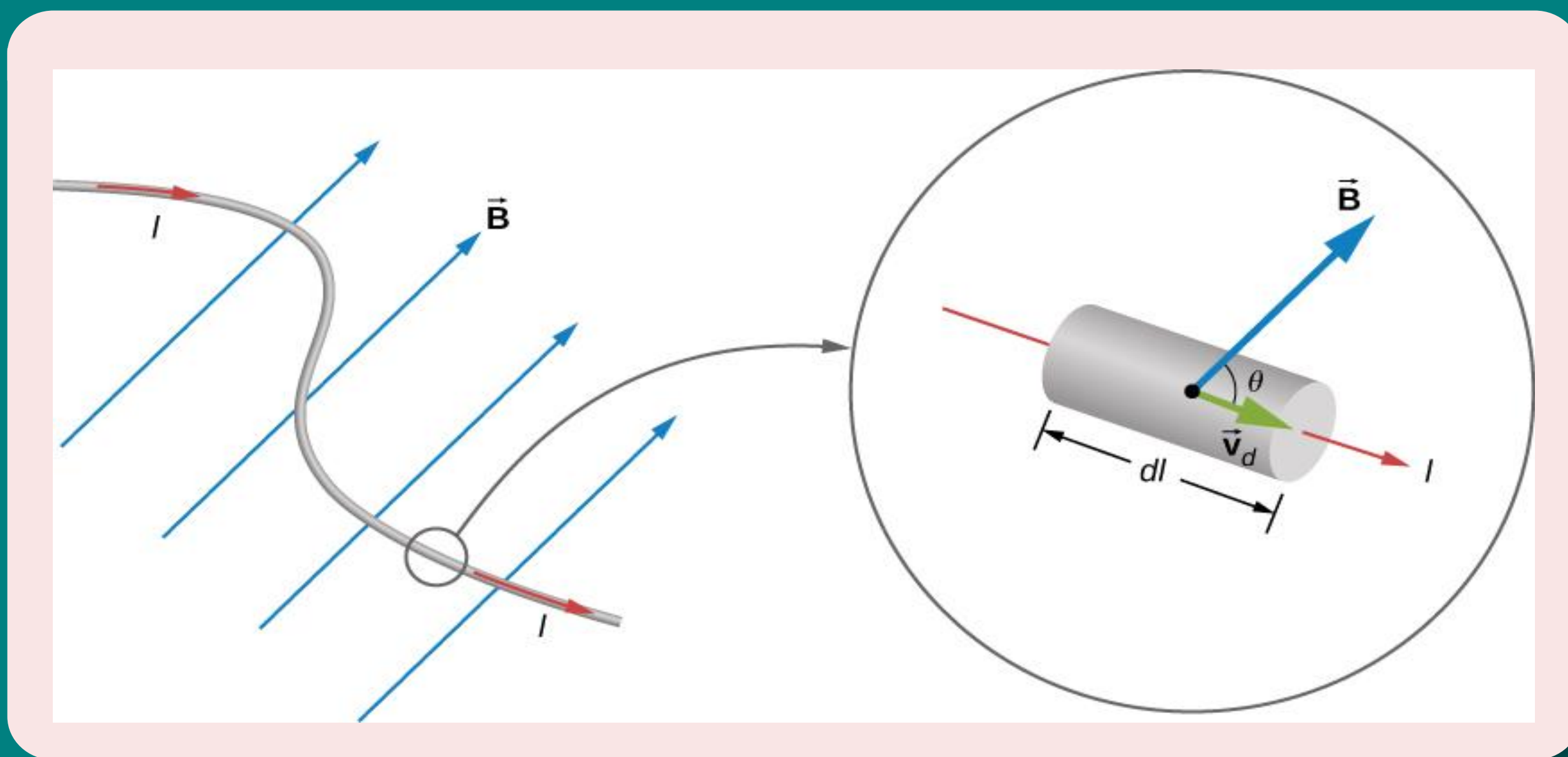
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$$d\vec{F} = Q\vec{v}_d \times \vec{B}$$
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$$d\vec{F} = neAv_d d\vec{l} \times \vec{B}$$

Magnetic force on a current-carrying conductor



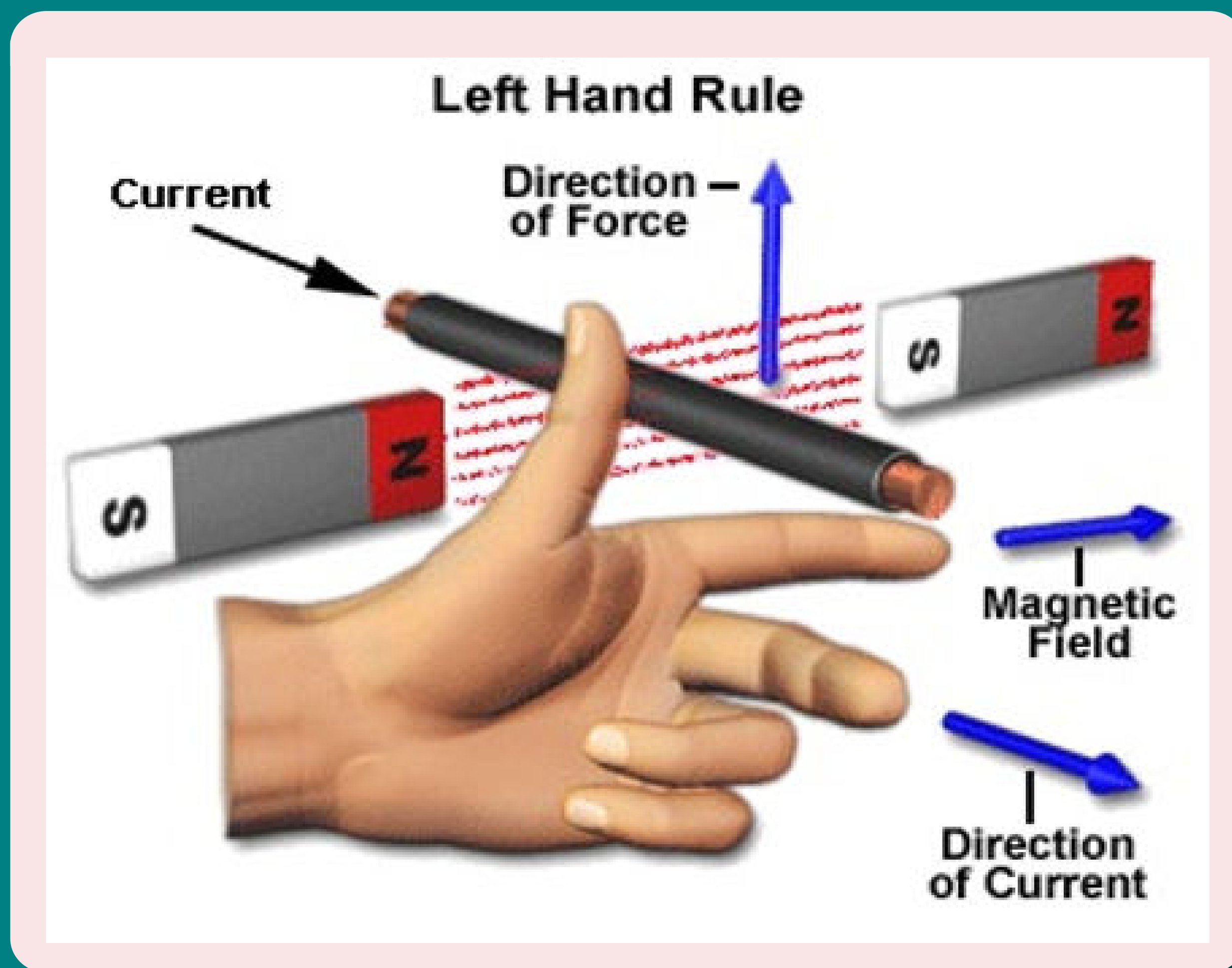
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$$d\vec{F} = neAv_d d\vec{l} \times \vec{B}$$
$$\boxed{d\vec{F} = I d\vec{l} \times \vec{B}} \quad (\because I = neAv_d)$$

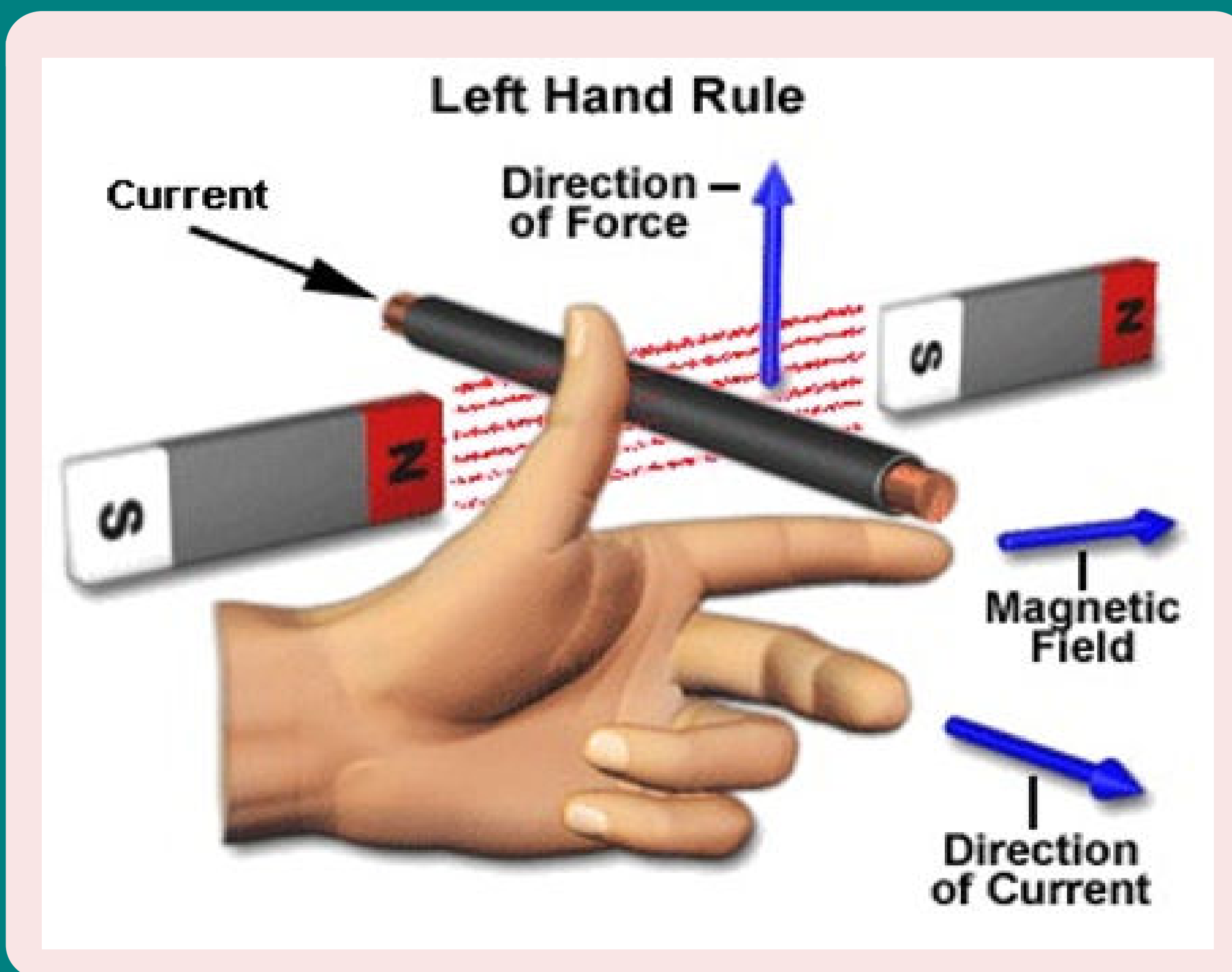
Magnetic force on a current-carrying conductor



- The force on a straight, current-carrying wire in a uniform magnetic field.

$$\vec{F} = I\vec{l} \times \vec{B}$$

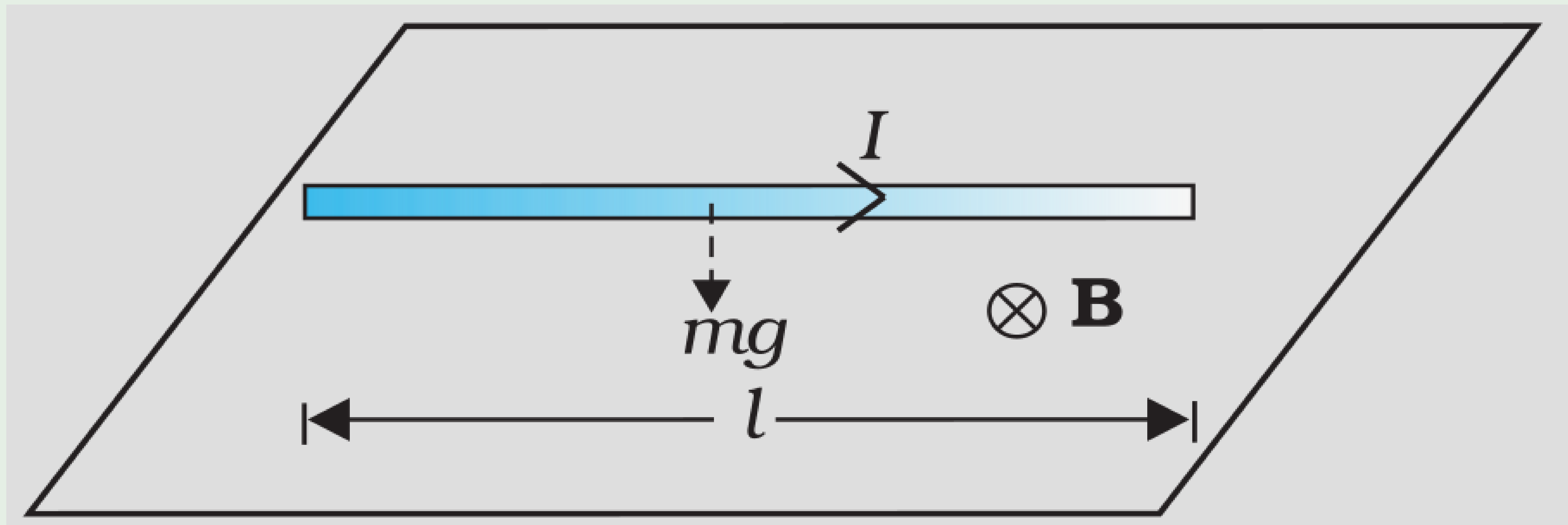
Magnetic force on a current-carrying conductor



- The force on a straight, current-carrying wire in a uniform magnetic field.
- $$\vec{F} = I\vec{l} \times \vec{B}$$
- The direction of this force is given by Fleming's Left hand rule.

Example 4.1

A straight wire of mass 200 g and length 1.5 m carries a current of 2 A . It is suspended in mid-air by a uniform horizontal magnetic field B . What is the magnitude of the magnetic field?



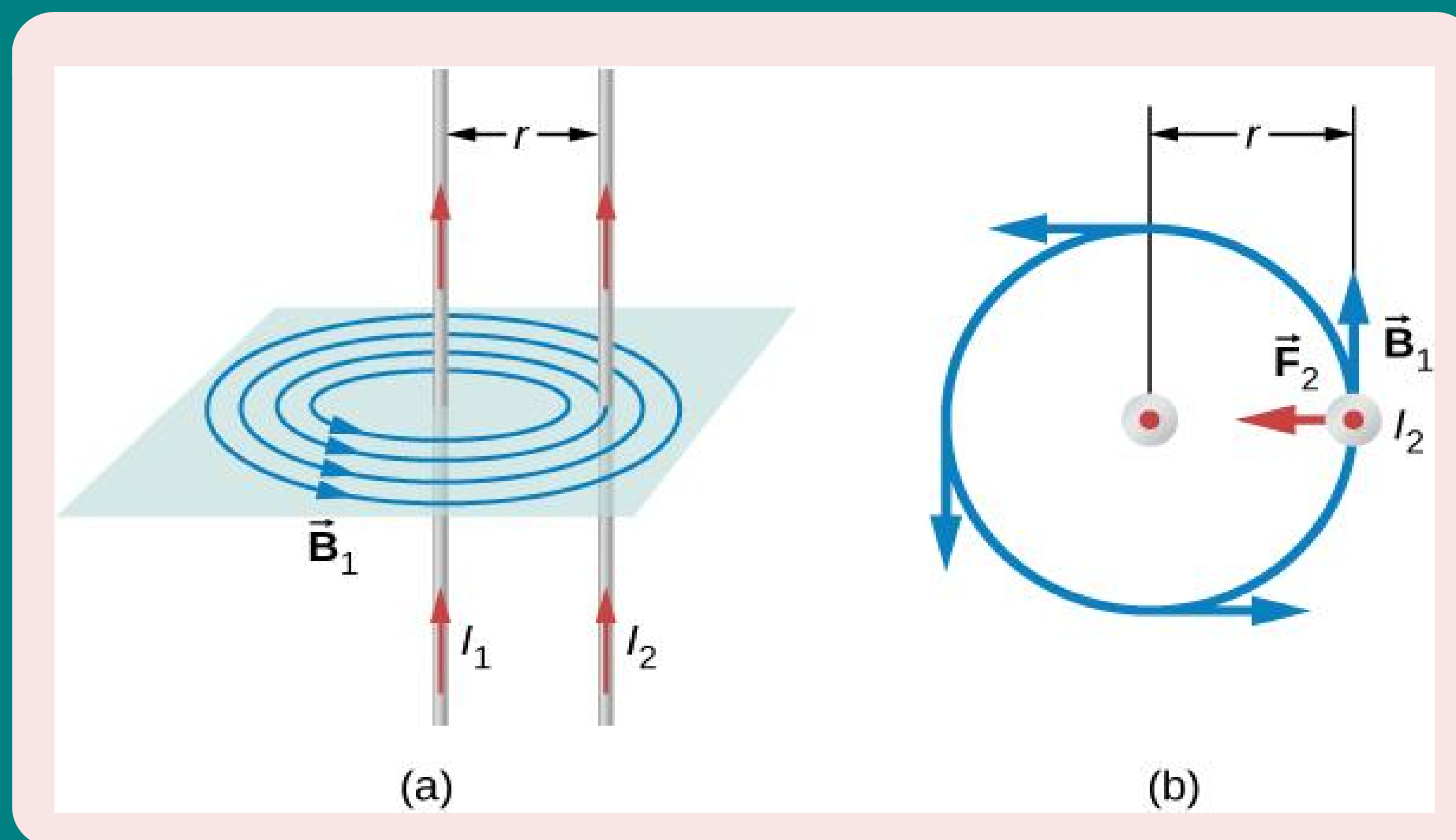
Exercise 4.5

What is the magnitude of magnetic force per unit length on a wire carrying a current of 8 A and making an angle of 30° with the direction of a uniform magnetic field of 0.15 T ?

Exercise 4.6

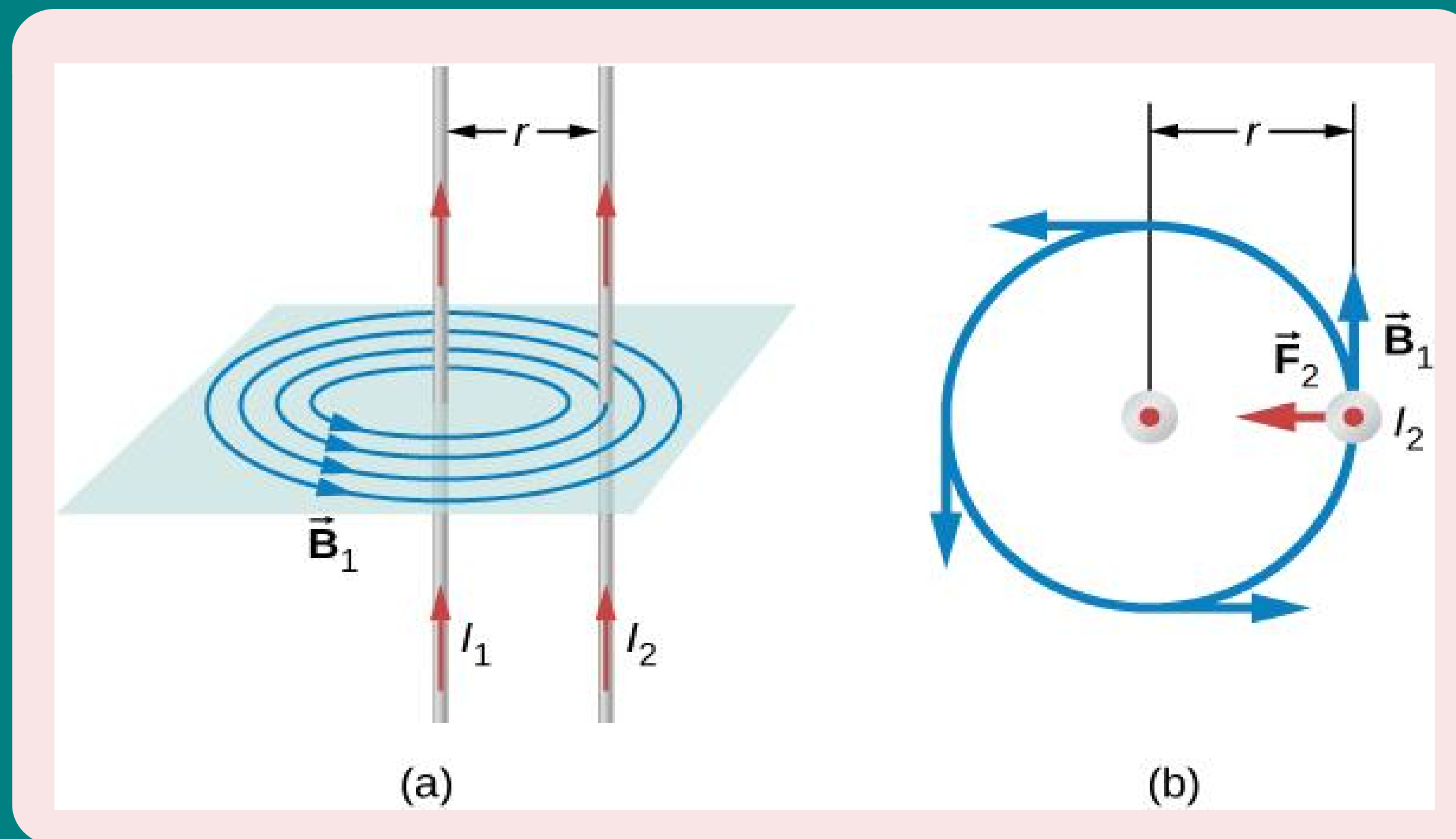
A 3.0 cm wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. The magnetic field inside the solenoid is given to be 0.27 T. What is the magnetic force on the wire?

FORCE BETWEEN TWO PARALLEL CURRENTS



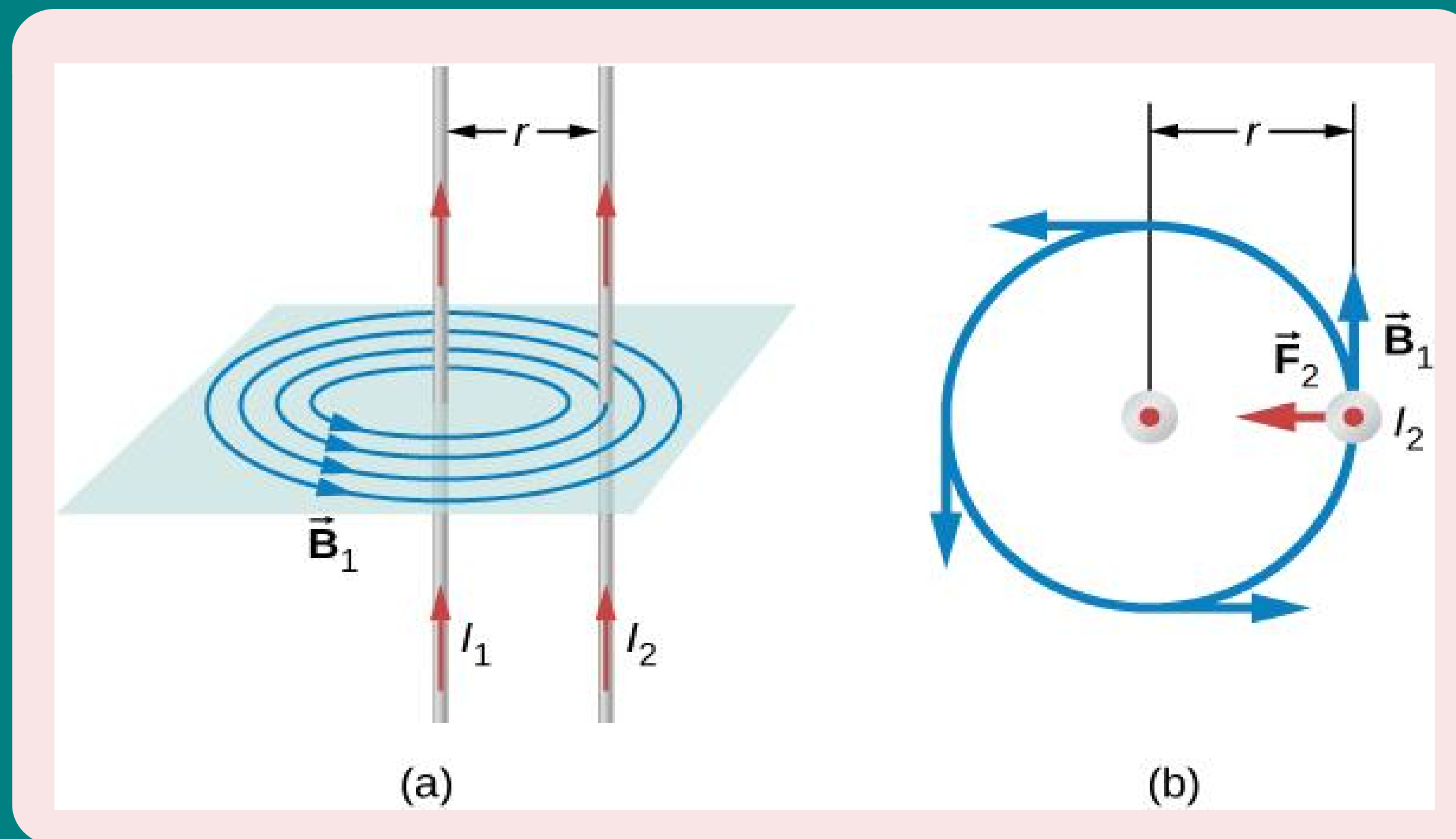
- We know that a **magnetic field is generated by a current-carrying conductor.**

FORCE BETWEEN TWO PARALLEL CURRENTS



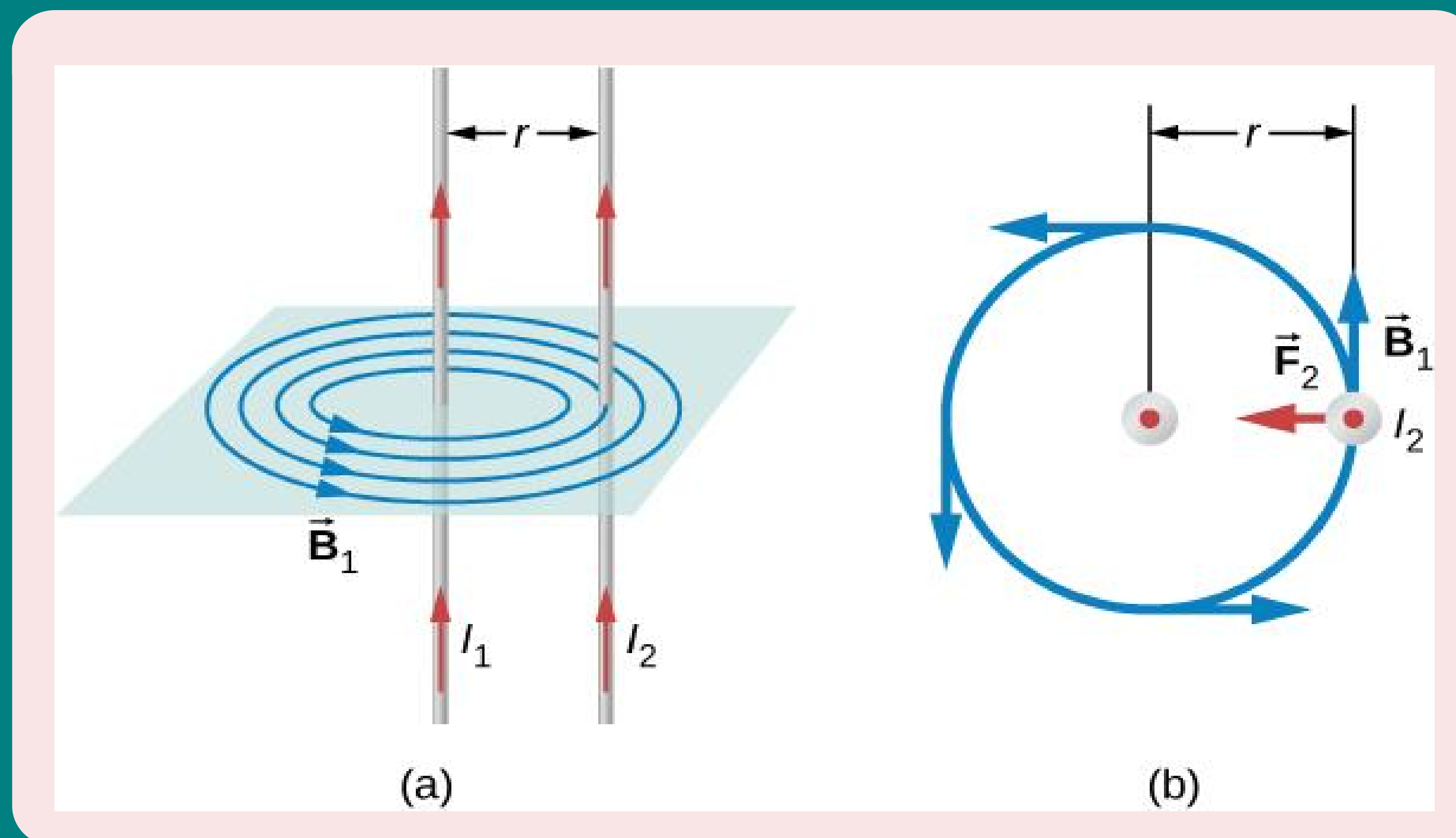
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- **Another current-carrying conductor experiences force** as a result of the **external magnetic field**.

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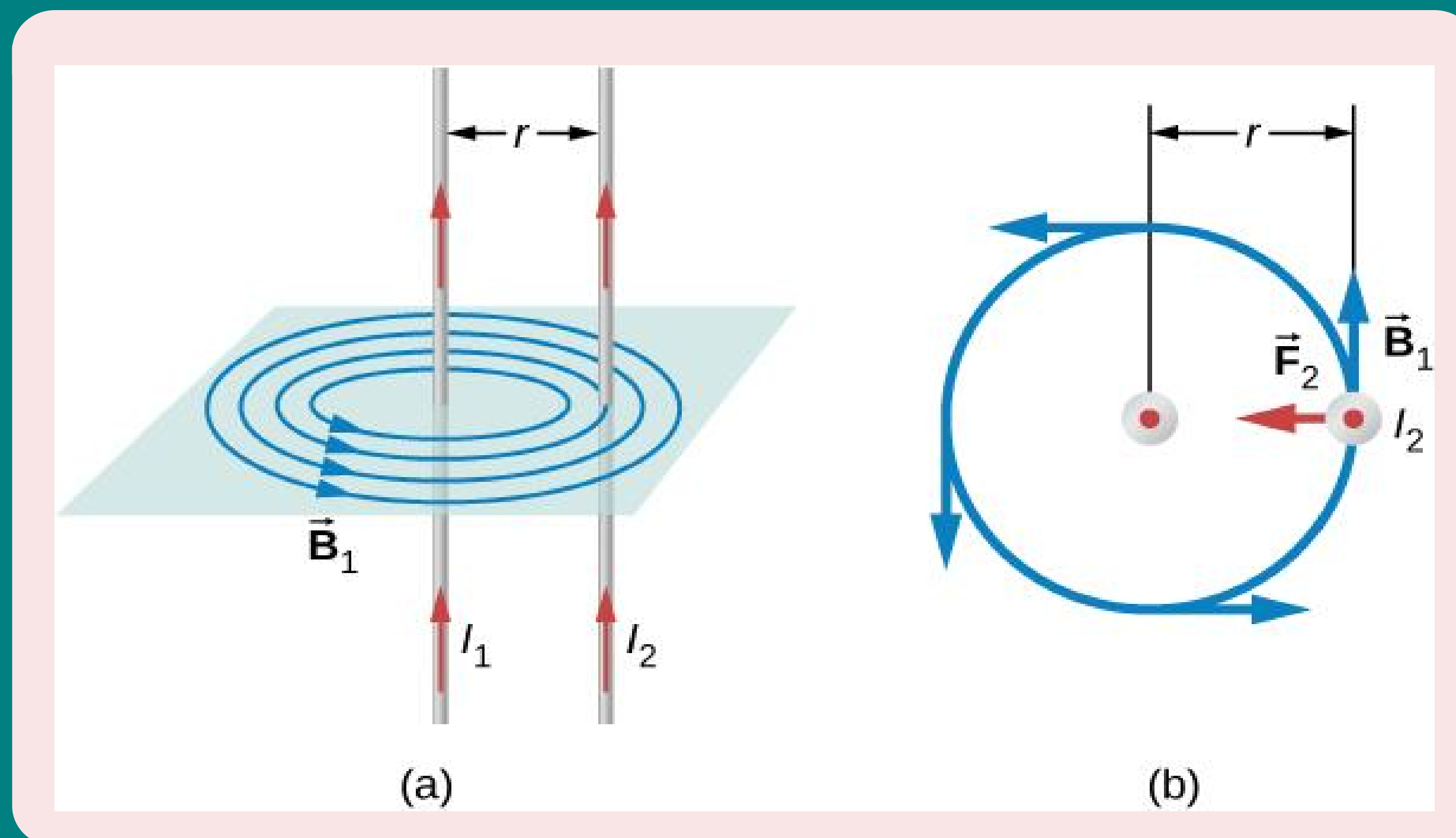
- We know that a **magnetic field** is generated by a **current-carrying conductor**.
- **Another current-carrying conductor experiences force** as a result of the **external magnetic field**.
- Consider **two long parallel current-carrying wires**, separated by a **distance 'r'**, such that **one of the wires is carrying current I_1** and the **other is carrying I_2** .

FORCE BETWEEN TWO PARALLEL CURRENTS



- The magnitude of the field due to I_1 at a distance r is

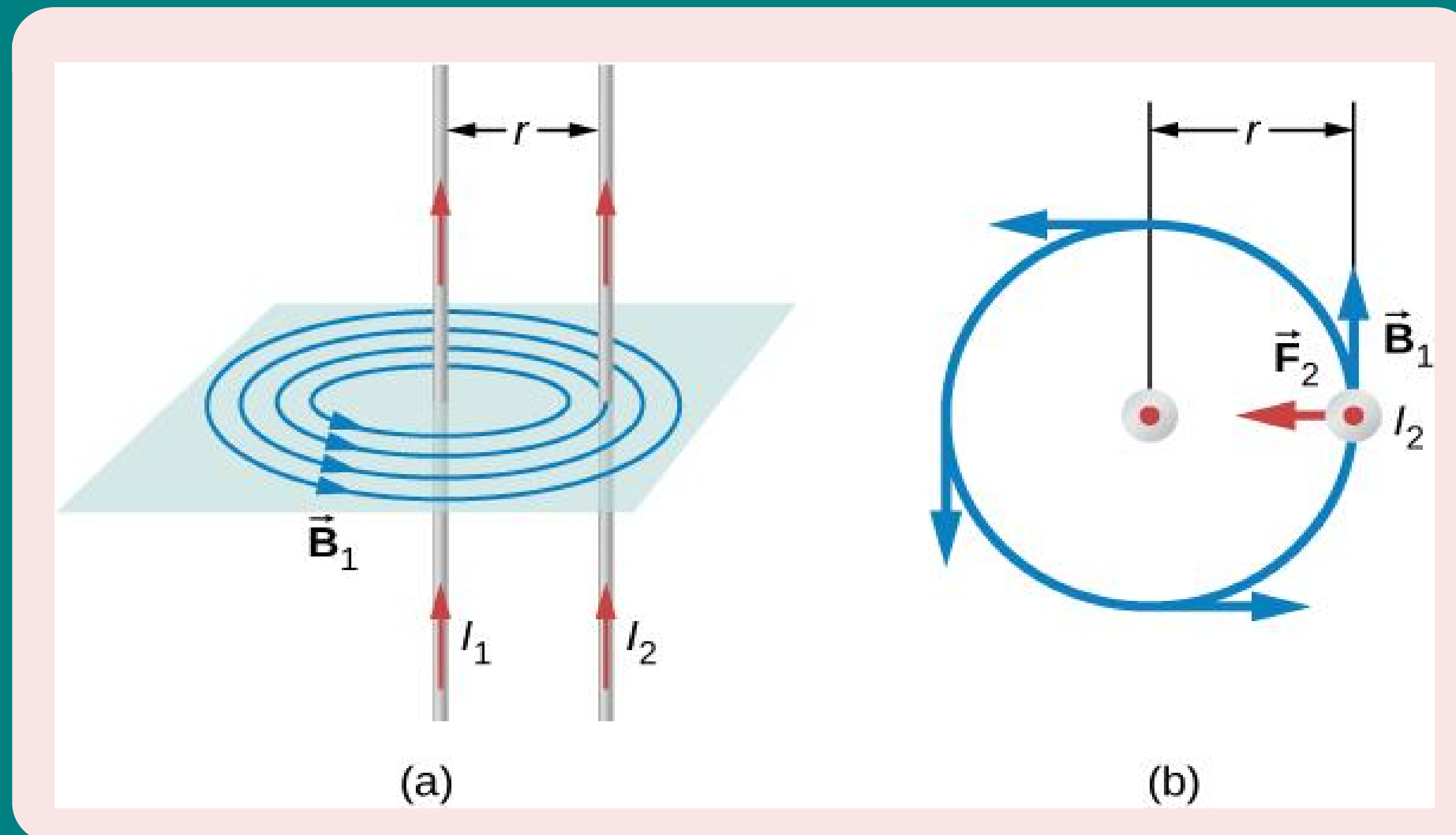
FORCE BETWEEN TWO PARALLEL CURRENTS



- The magnitude of the field due to I_1 at a distance r is

$$B_1 = \frac{\mu_0 I_1}{2\pi r} \dots (1)$$

FORCE BETWEEN TWO PARALLEL CURRENTS

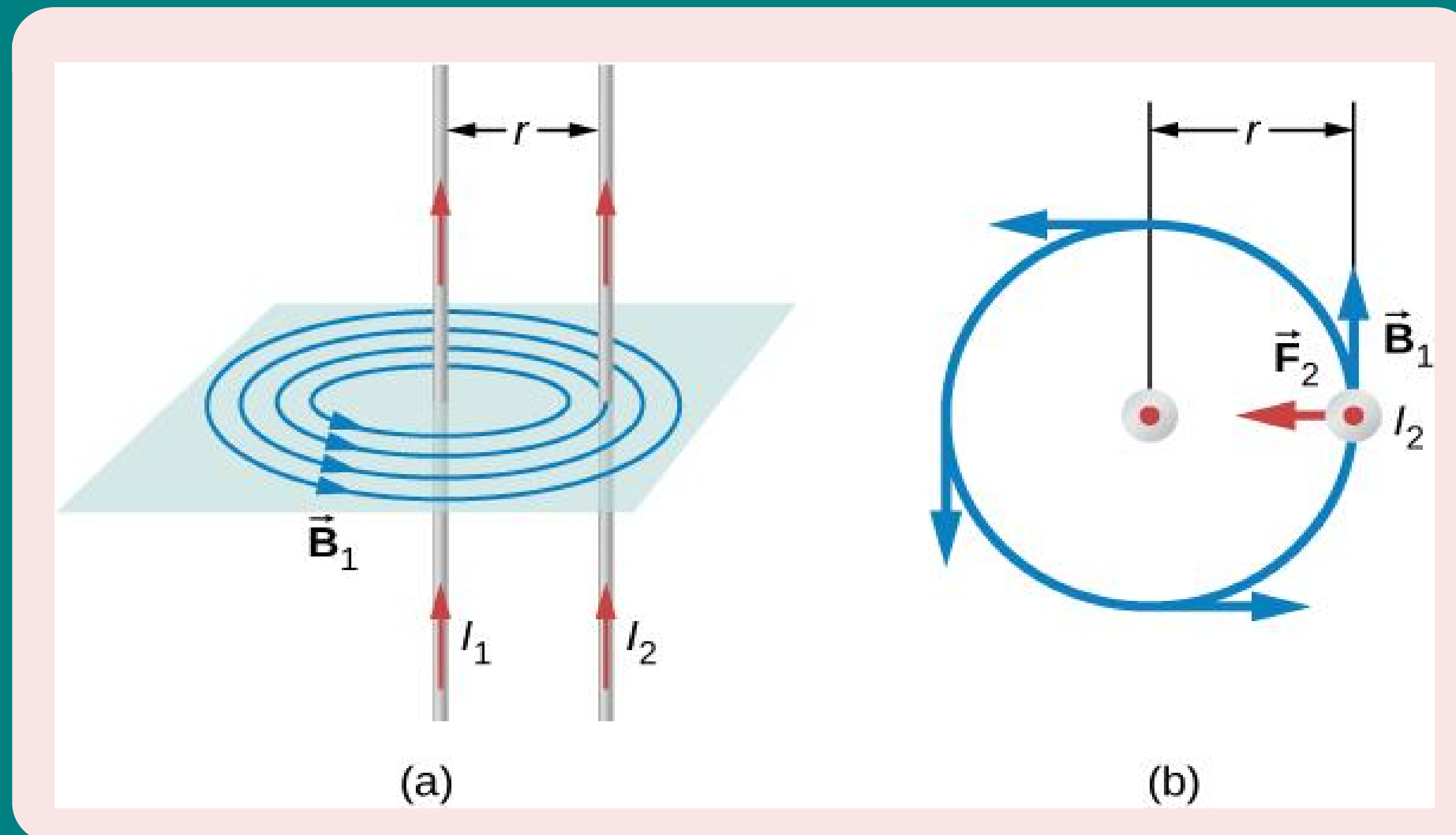


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- This **field (B_1)** produced by wire 1 **exert a force (F_{21}) on a length l of wire 2** and is given by.

FORCE BETWEEN TWO PARALLEL CURRENTS



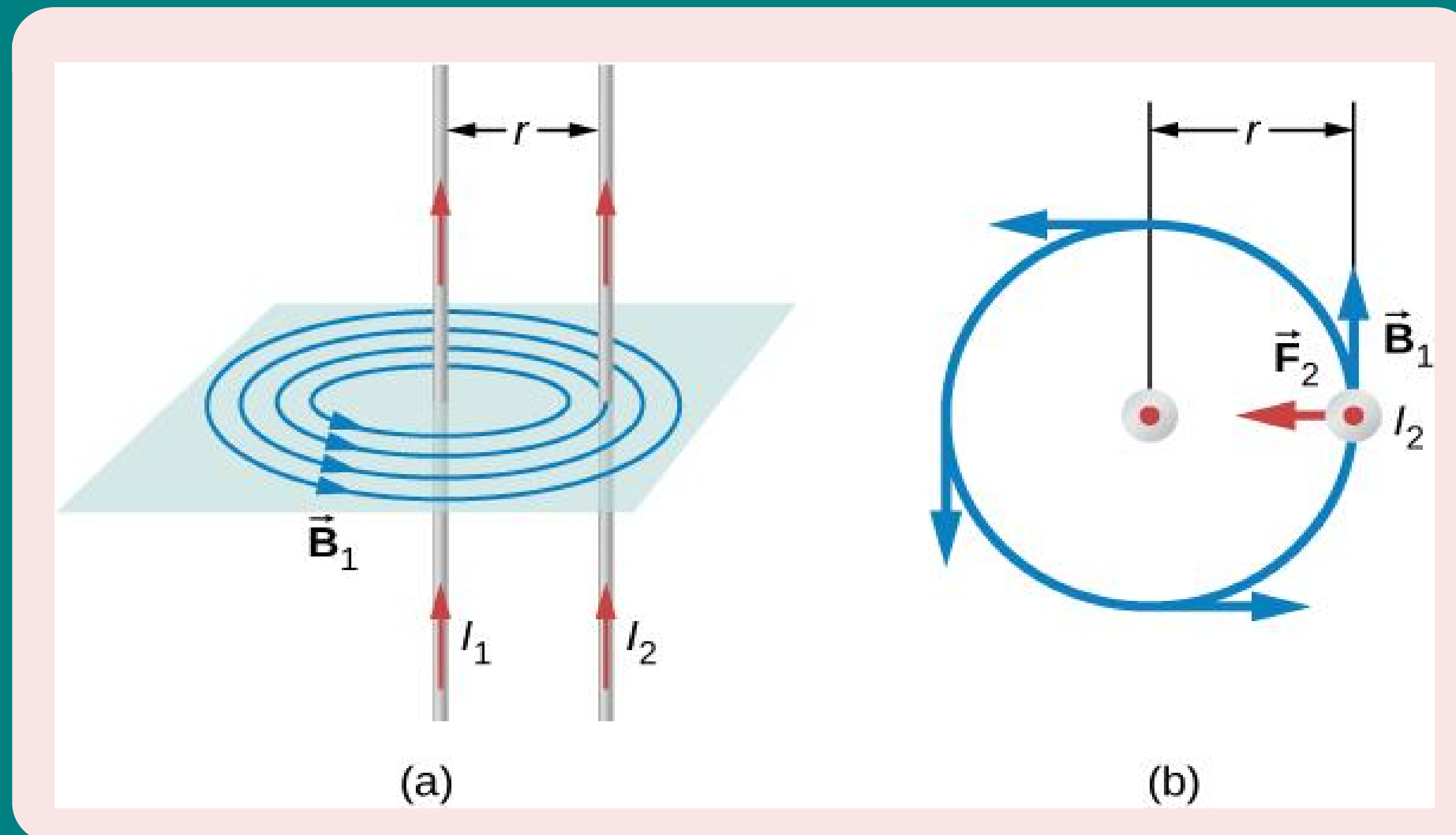
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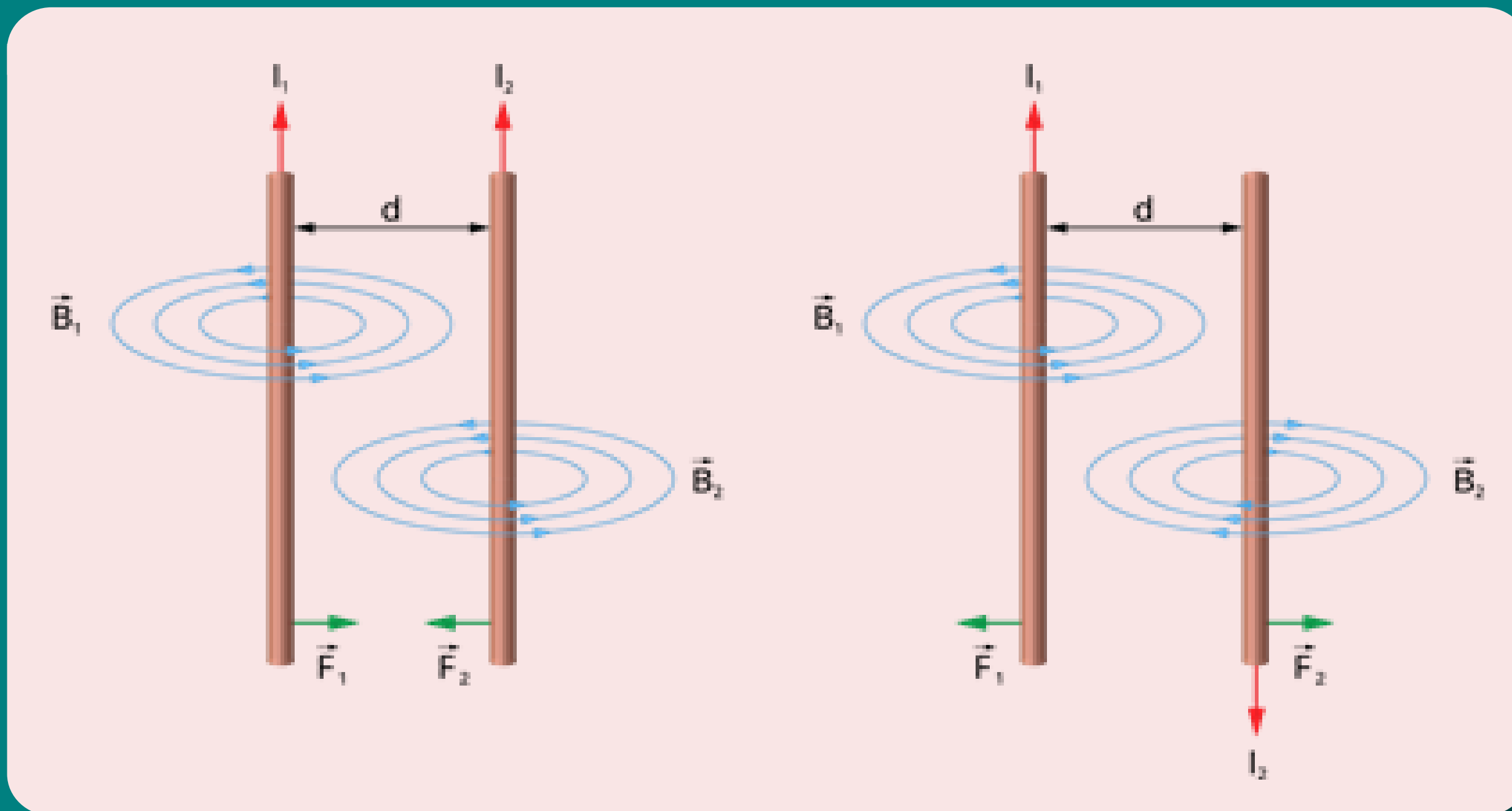
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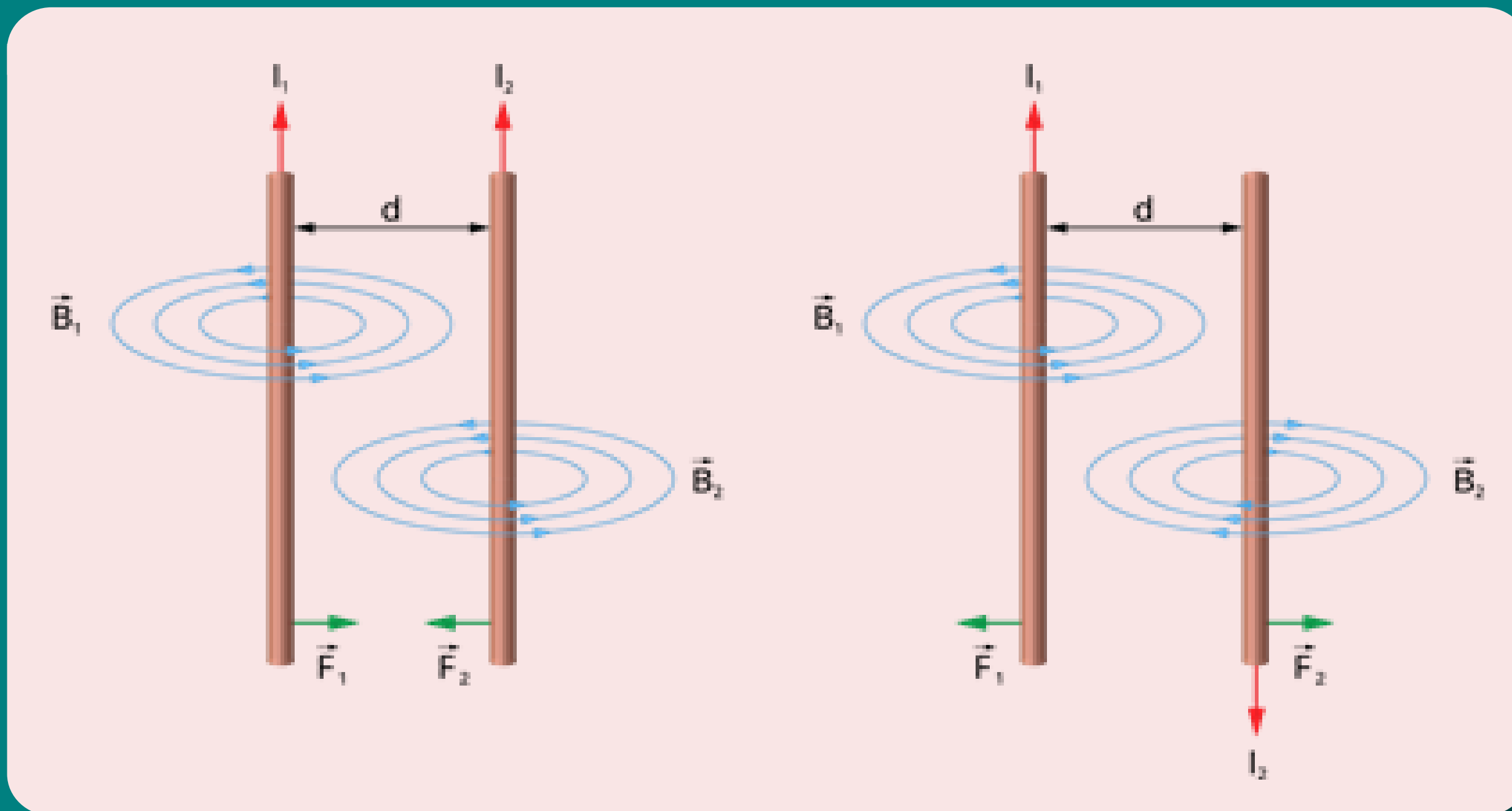
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FORCE BETWEEN TWO PARALLEL CURRENTS



- Similarly, **field** (B_2) produced by wire 2 **exert a force** (F_{12}) **on a length** l **of wire 1** and its direction is opposite to F_{21}

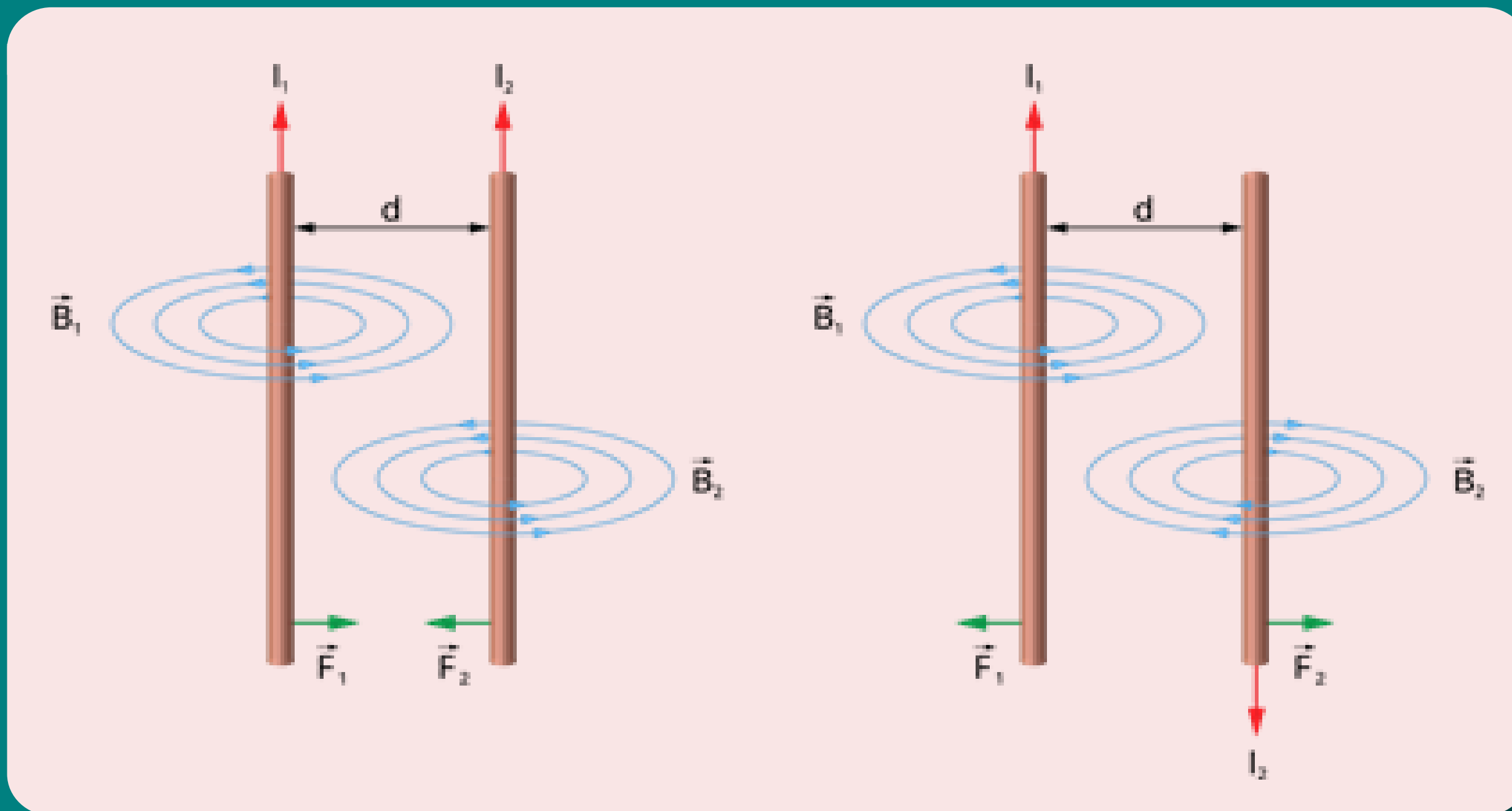
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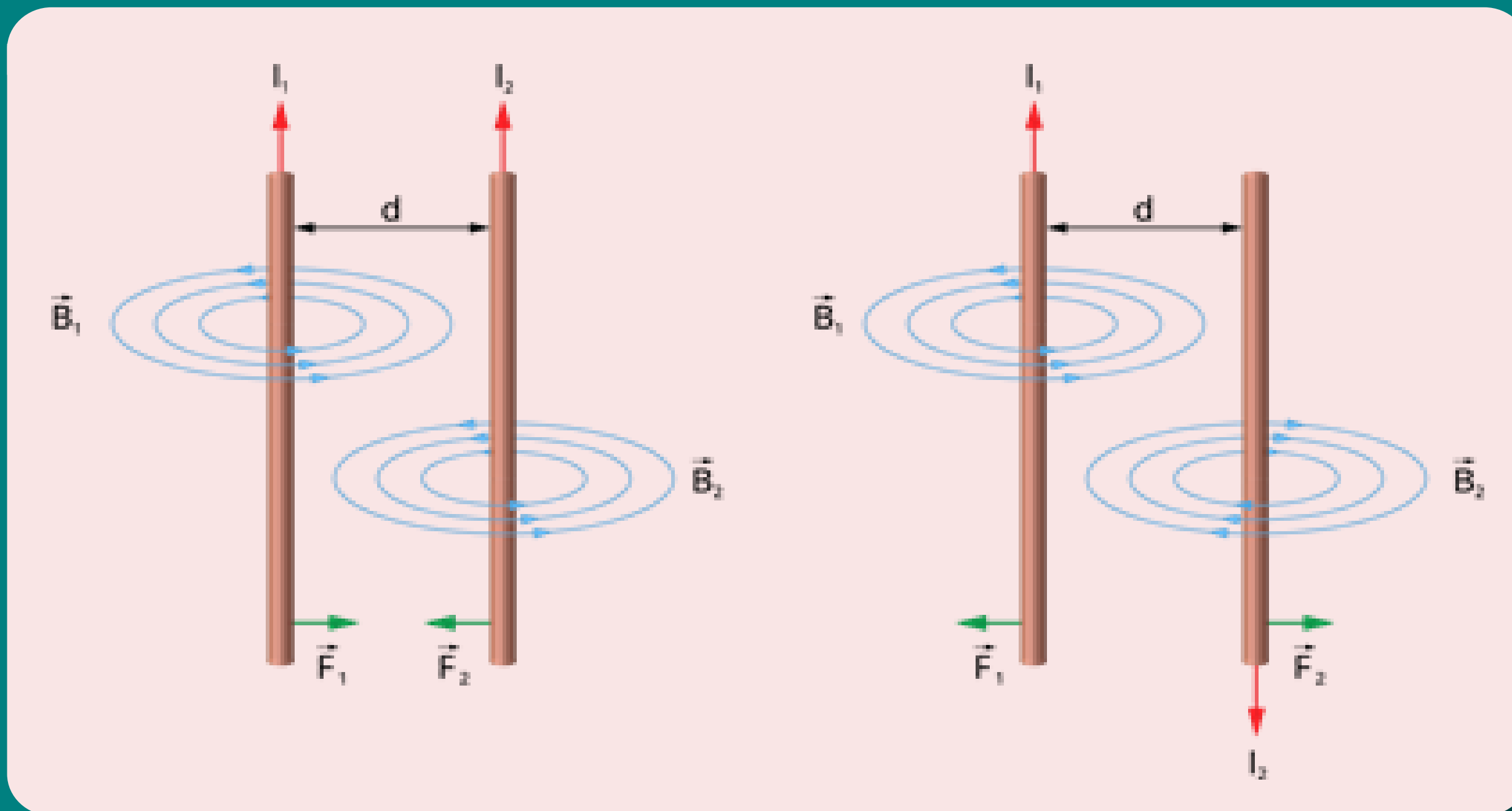
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FORCE BETWEEN TWO PARALLEL CURRENTS

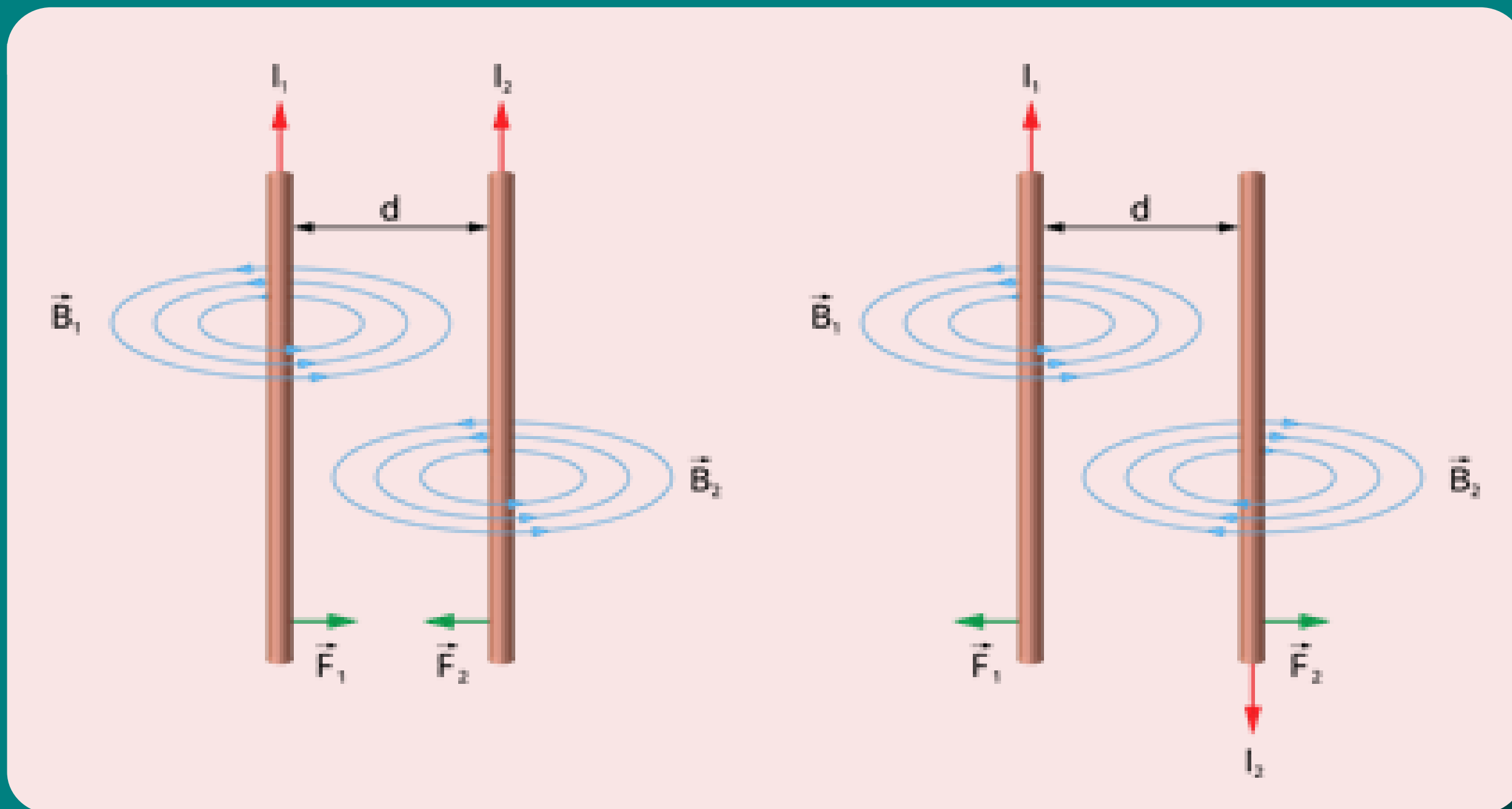


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- **Parallel currents (same direction) attract, and antiparallel currents (opposite direction) repel.**

FORCE BETWEEN TWO PARALLEL CURRENTS

Definition of S.I. Unit of current, Ampere

- The **definition of the ampere is based on the force between current - carrying wires**. Note that for long, parallel wires separated by **1 meter** with each carrying **1 ampere**, the **force per meter** is

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$$\frac{F}{l} = \left(10^{-7} \frac{T \cdot m}{A} \right) \frac{2 A^2}{1 m} = 2 \times 10^{-7} N/m$$

Exercise 4.7

Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A.